

The Art of Value: Pricing Strategy & Marketing

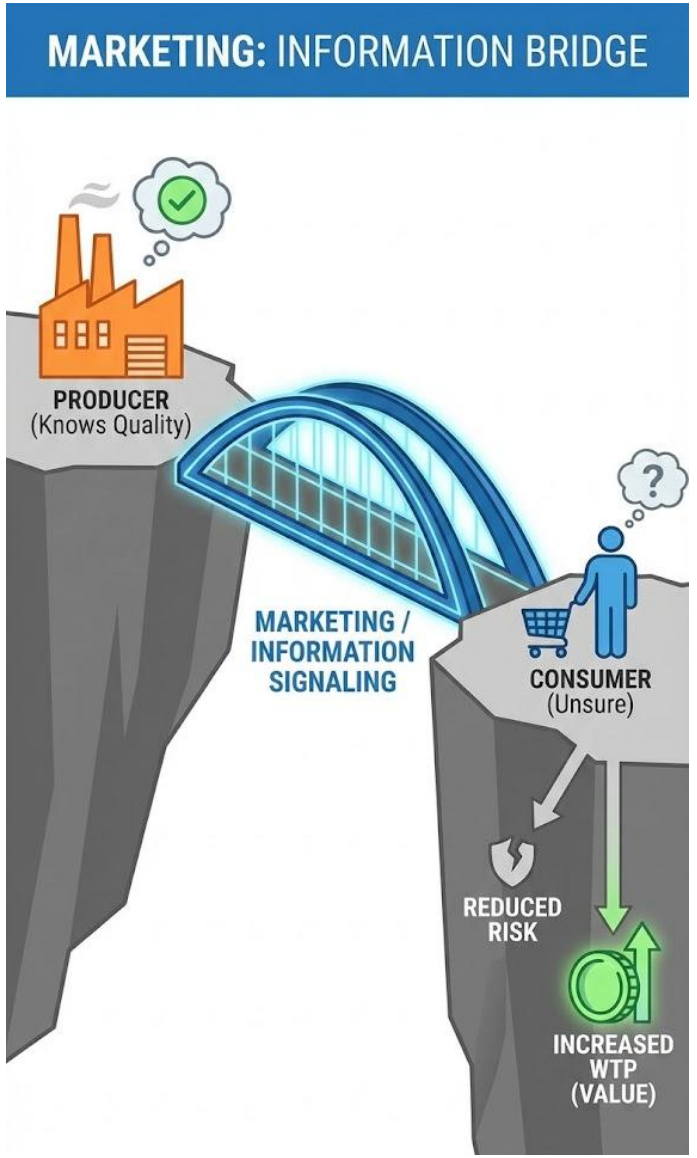
Marketing, Pricing Algorithms, and Value Capture

ITGE275 Economics

Associate Prof. Dr. Suppawong Tuarob
Faculty of ICT, Mahidol University

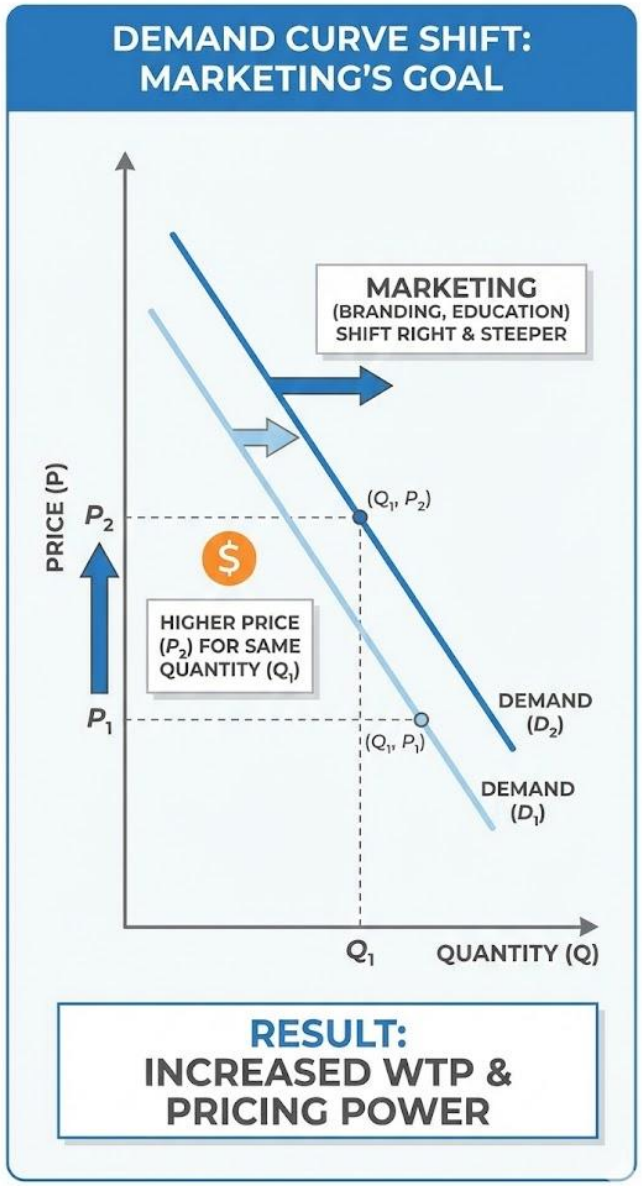
Marketing != Just Advertising

- **Common Myth:** Marketing is just making cool logos and ads.
- **Economic Reality:** Marketing is about **Information** and **Value Signaling**.
- **The Problem:** Information Asymmetry.
  - Producer knows the product is good.
  - Consumer isn't sure.
- **The Solution:** Marketing bridges this gap, reducing the risk for the buyer and increasing their Willingness To Pay (WTP).



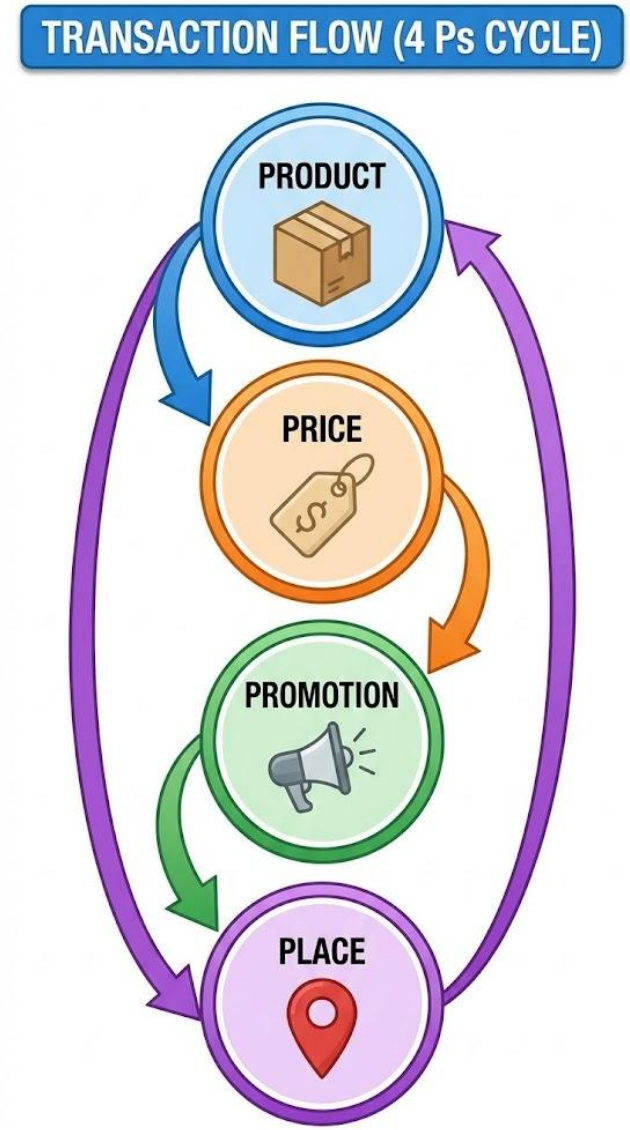
The Demand Curve Shift

- **Recall:** The Demand Curve represents WTP.
- **Marketing's Goal:** To shift the Demand Curve to the **Right** (and make it steeper/inelastic).
- **Mechanism:**
  - **Branding:** "This isn't just coffee; it's Starbucks." (Inelasticity).
  - **Education:** "You need this vitamin to live longer." (Shift Right).
- **Result:** You can charge a higher price (P_2) for the same quantity (Q_1).



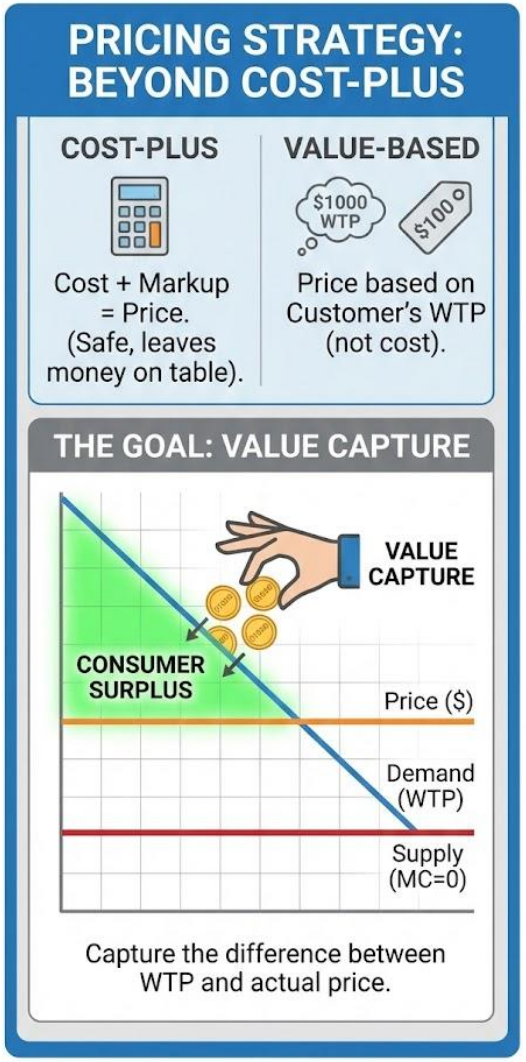
The Business Transaction Flow

- Every transaction is a contract.
- **Components:**
  - **Product:** The good/service.
  - **Price:** The capture of value.
  - **Promotion:** The signal sent to the buyer.
  - **Place:** The distribution channel (App, Store).
- **The "4 Ps" of Marketing:** Product, Price, Promotion, Place.



Pricing Strategy: Beyond Cost-Plus

- **Cost-Plus Pricing:** Cost + Markup = Price. (Safe, but leaves money on the table).
- **Value-Based Pricing:** Price based on the **Customer's WTP**, not your cost.
  - *Example:* Software costs $0 to copy (MC=0), but you charge $100 because it saves the user $1000 in time.
- **The Goal:** Capture the **Consumer Surplus** (The difference between what they were willing to pay and what they actually paid).



Psychological Pricing

- Humans are not perfectly rational algorithms (Behavioral Economics).
- **Left-Digit Bias:** $9.99 feels significantly cheaper than $10.00.
- **Anchoring:** Placing a $2000 watch next to a $500 watch makes the $500 watch seem like a "bargain."
- **Decoy Effect:** Small Popcorn ($3), Medium ($6.50), Large ($7). The Medium exists only to push you to the Large.



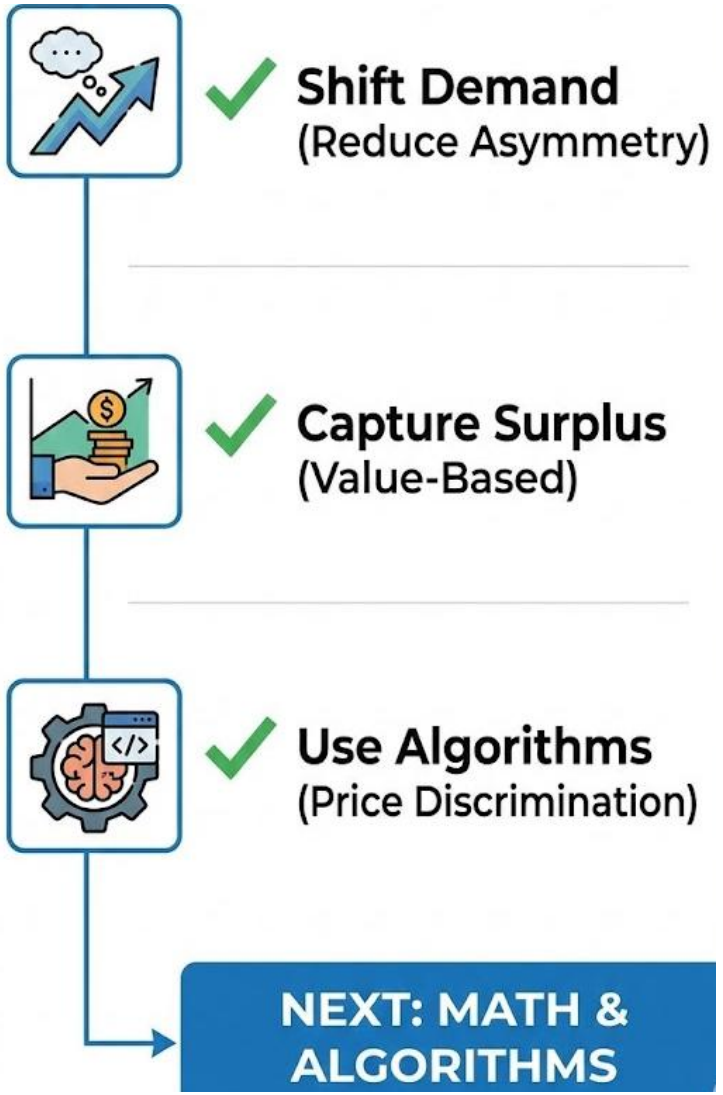
Introduction to Algorithmic Pricing

- In the digital age, prices are not written in ink; they are written in code.
- **Static Pricing:** Same price for everyone, forever (e.g., printed menu).
- **Dynamic Pricing:** Price changes based on time, demand, and user profile (e.g., Uber, Amazon, Agoda).
- **Advantages:** Allows firms to clear inventory and maximize revenue efficiently.



Part 1 Summary

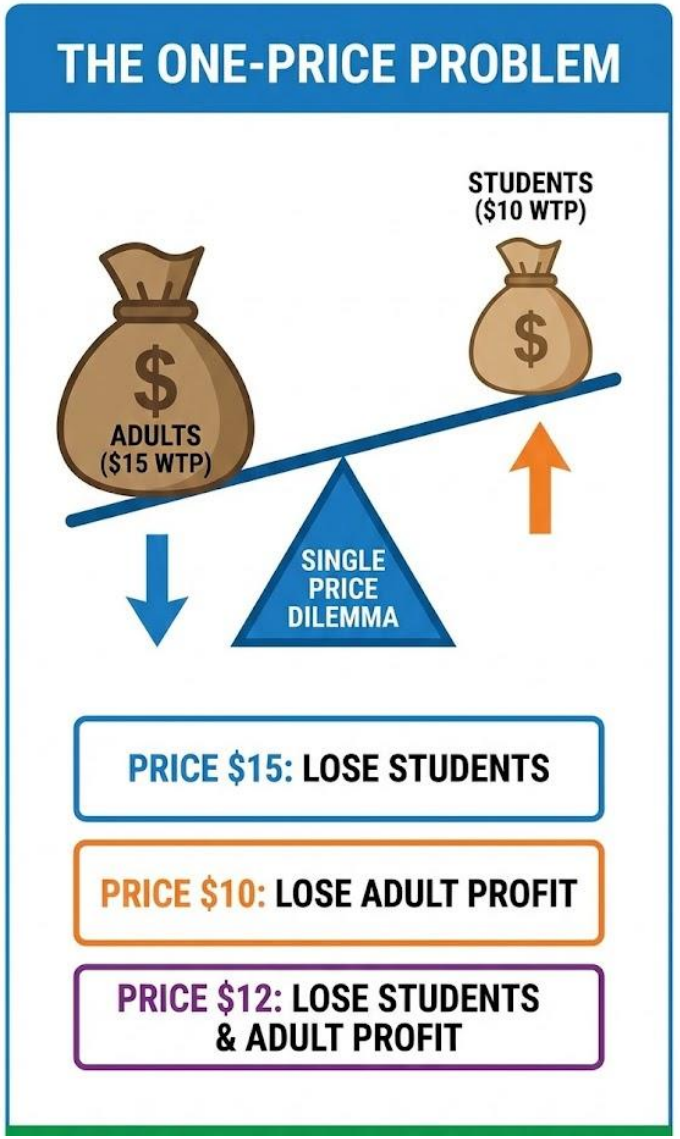
- Marketing shifts demand by reducing information asymmetry.
- Value-Based Pricing > Cost-Plus Pricing.
- Psychology matters (99 cents).
- **Next:** How to use Math to charge different people different prices (Price Discrimination).



Part 2: Price Discrimination

The One-Price Problem

- **Scenario:** You run a Cinema.
  - **Adults:** Willing to pay $15.
  - **Students:** Willing to pay $10.
- **Dilemma:**
  - If Price = $15: You lose all Students.
  - If Price = $10: You lose $5 profit from every Adult.
  - If Price = $12: You lose Students AND lose $3 from Adults.
- **Solution:** Charge two different prices.



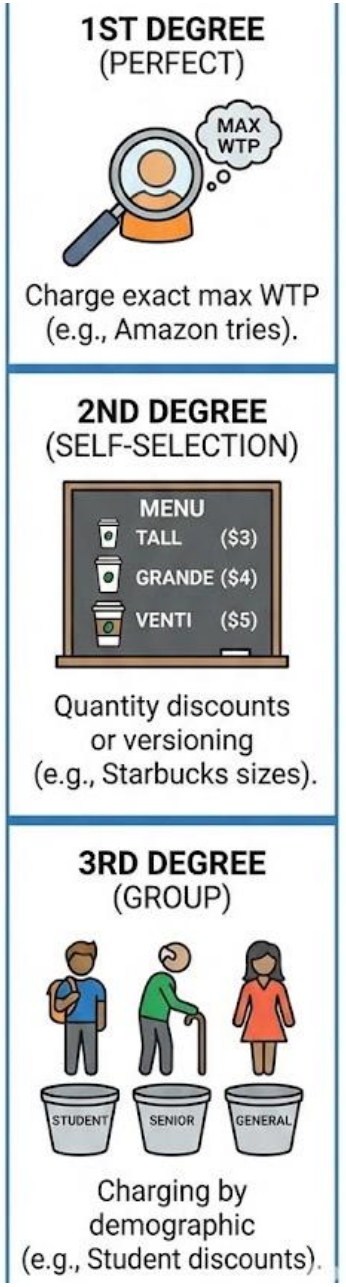
Price Discrimination Defined

- **Definition:** Selling the exact same product to different buyers at different prices.
- **Requirements:**
  - **Market Power:** You must be a price maker.
  - **Segmentation:** You must be able to identify who is who (ID cards, Cookies).
  - **No Arbitrage:** Students shouldn't be able to resell their tickets to Adults.



Types of Price Discrimination

- **1st Degree (Perfect):** Charge every single person their exact max WTP. (Rare, but Amazon tries).
- **2nd Degree (Self-Selection):** Quantity discounts or versioning.
  - *Example:* Starbucks Tall vs. Venti. Airline Economy vs. Business. (You choose your tier).
- **3rd Degree (Group):** Charging by demographic.
  - *Example:* Student discounts, Senior Citizen rates, Ladies' Night.

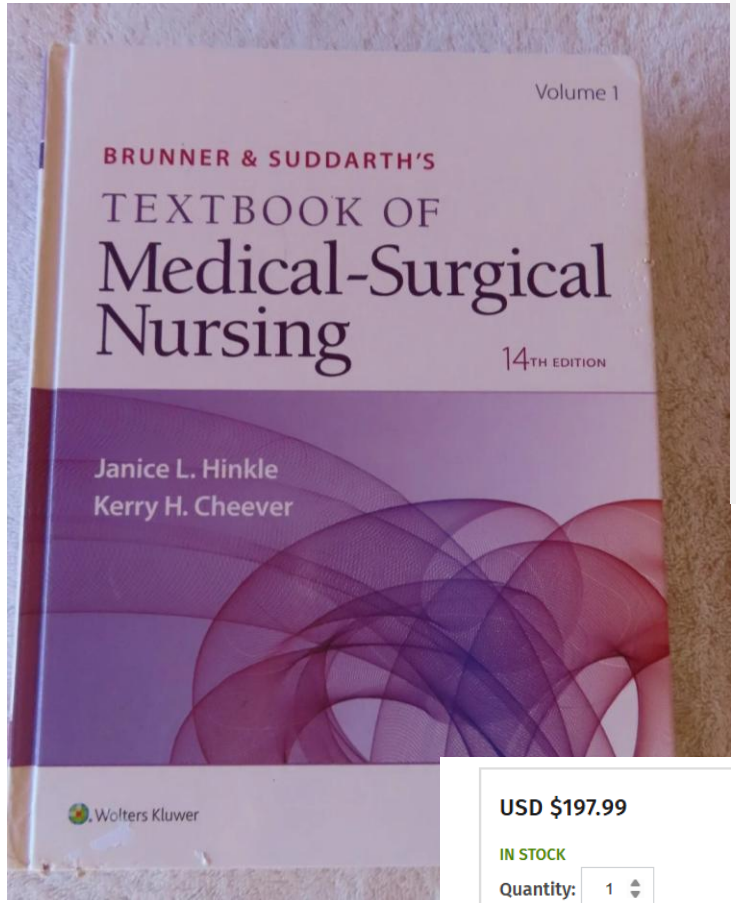


Example


อัตราค่าบริการผ่านเข้าอุทยานแห่งชาติ
อุทยานแห่งชาติทองผาภูมิ จังหวัดกาญจนบุรี
Entrance Fee Rate For Thong Pha Phum National Park
ที่ | รายการ | อัตราค่าบริการ (บาท/คน/คัน/คืน)
1. ค่าบริการ สำหรับบุคคล Personal Fee
1.1 ชาวไทย Thai Fee
ผู้ใหญ่ Adult 40-
เด็ก Children 20-
1.2 ชาวต่างประเทศ Foreigner Fee
ผู้ใหญ่ Adult 200-
เด็ก Children 100-


เริ่ม 1 กุมภาพันธ์ 2558
อัตราค่าบริการในการใช้พื้นที่ในเขตอุทยานแห่งชาติแก่งกระจาน
Entrance Fee to The Park
ชาวไทย
ผู้ใหญ่ 100 บาท
เด็ก 40 บาท
Foreigner
Adults 300 ฿ (Each)
Child 200 ฿ (Each)
อัตราค่าพักแรม 30 บาท/คน/คืน
อัตราค่ารถยนต์ 30 บาท/คัน
Vehicle 30 ฿
Tent (Overnight) 30 ฿ Per Person Person Per Night

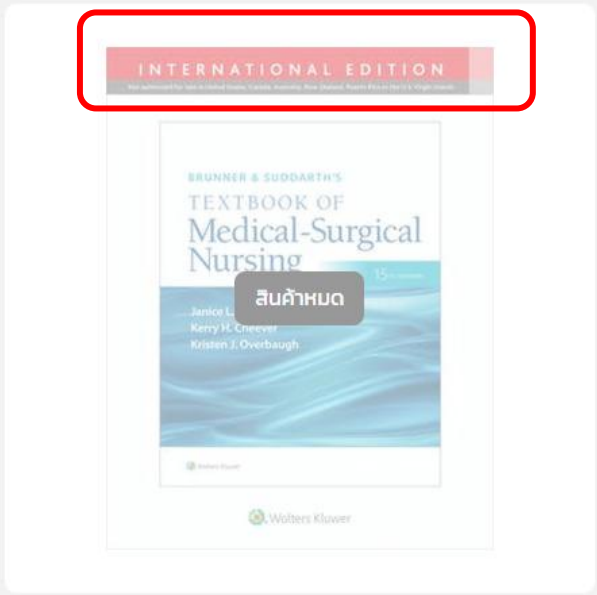
Example



USD $197.99

IN STOCK

Quantity: 1





BRUNNER & SUDDARTH'S TEXTBOOK OF MEDICAL-SURGICAL NURSING (IE) (HC)

ผู้แต่ง : JANICE L. HINKLE et al.

อ่านรีวิว เพิ่มในรายการโปรด

หนังสือปกแข็ง

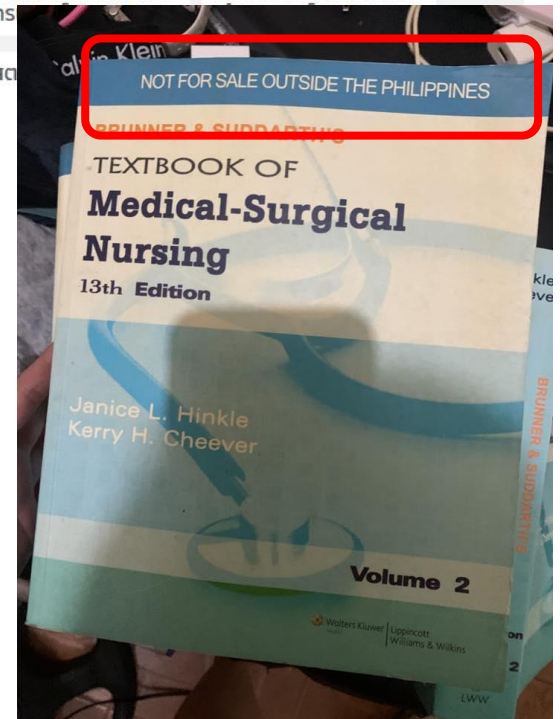
฿ 3,744.00 ราคาปกติ ~~4,160.00~~ ประหยัด 10 %

คูปองที่ใช้ได้ (กรอกโค้ดในขั้นตอนการจ่ายเงินเพื่อรับส่วนลดเพิ่มเติม)

Valen-Deal ซื้อครบ 299 ลด 14.- Valen-Deal ซื้อครบ 899 ลด 24.- Valen-Deal ซื้อครบ 1499 ลด 34.-

ซื้อทันที หยิบใส่รถเข็น

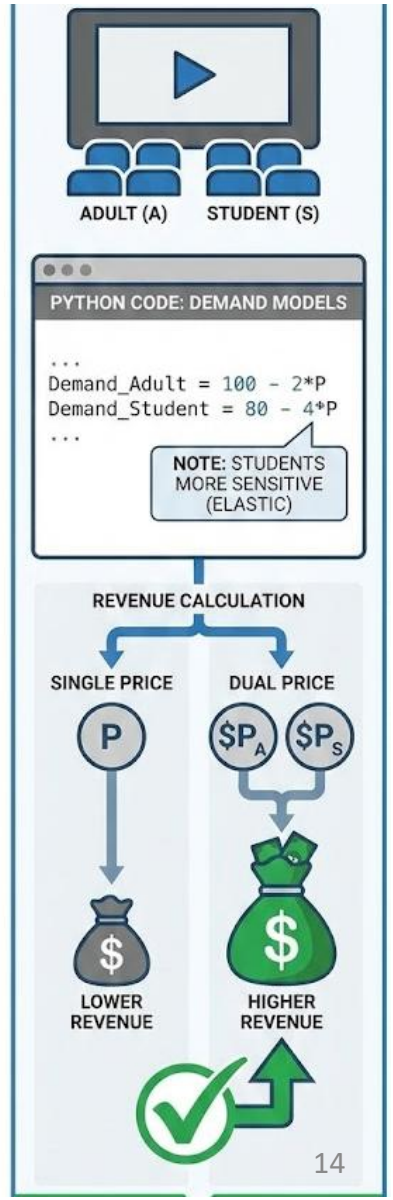
TAGS : การ พยาบาลศาสต



Original price: $197.99 ~
6,218.87 Thai Baht

Simulating Discrimination (Python Preview)

- In the Notebook, we model the Cinema scenario.
- **Code Logic:**
  - `Demand_Adult = 100 - 2*P`
  - `Demand_Student = 80 - 4*P` (Students are more price sensitive/elastic).
- **Calculation:**
  - We calculate `Revenue_Single_Price`.
  - We calculate `Revenue_Dual_Price`.
- **Result:** Dual Price always wins.



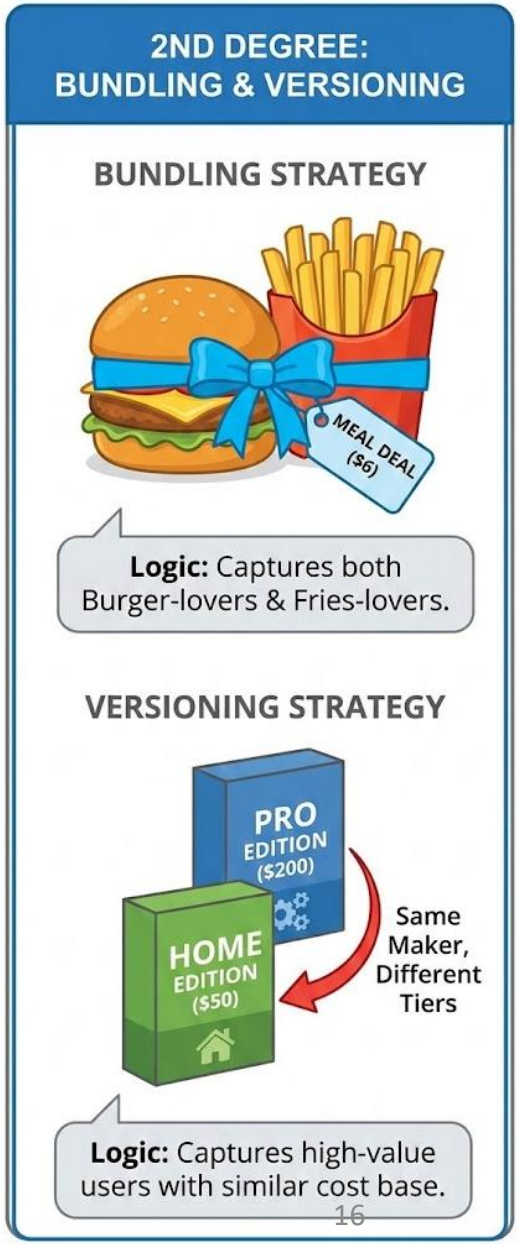
1st Degree Discrimination & Big Data

- In the past, Perfect Discrimination was impossible.
- **Now:** Companies use Big Data (Browser history, Location, Device type) to estimate your personal WTP.
- **Example:** Orbitz found Mac users spend 30% more on hotels, so they showed pricier options to Mac users.
- **Ethical Question:** Is this efficient or exploitative? (Captures all Consumer Surplus).



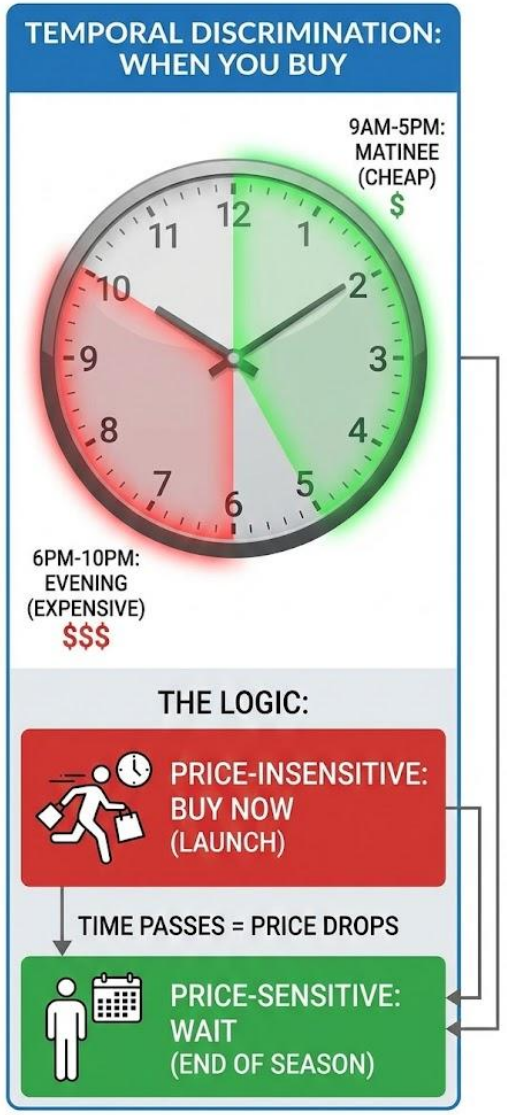
2nd Degree: The "Bundling" Strategy

- **Bundling:** Selling products together (Burger + Fries).
- **Logic:**
  - Person A loves Burgers ($5) but hates Fries ($1).
  - Person B hates Burgers ($1) but loves Fries ($3).
  - Bundle Price ($6) captures both.
- **Versioning:** Software "Pro" vs "Home" editions. The cost to make them is the same, but the price difference captures high-value corporate users.



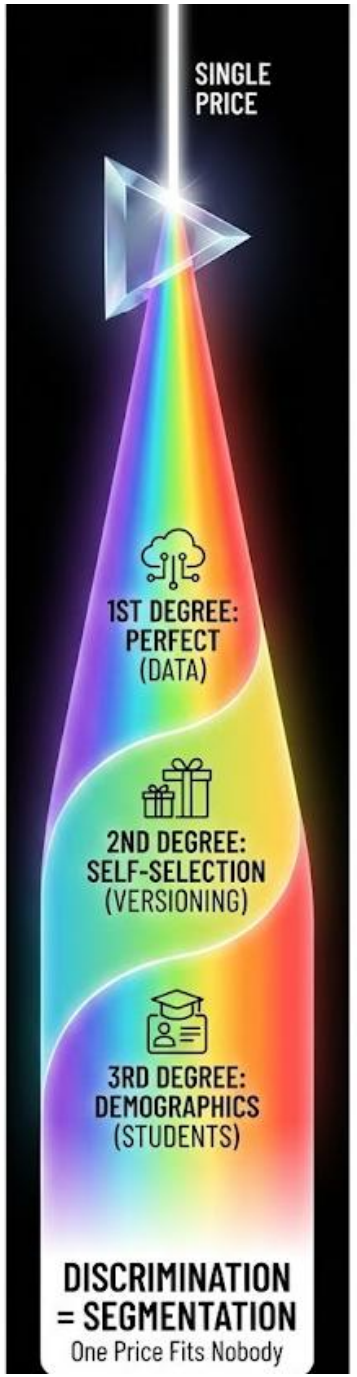
3rd Degree: Temporal Discrimination

- Charging based on **When** you buy.
- **Movies:** Matinee (Daytime) is cheaper than Evening.
- **Fashion:** Full price at launch → Discount at end of season.
- **Logic:** Price-insensitive people buy now. Price-sensitive people wait.



Part 2 Summary

- One price fits nobody.
- Discrimination = Segmentation.
- **1st Degree:** Perfect (Data-driven).
- **2nd Degree:** Self-selection (Versioning/Bundling).
- **3rd Degree:** Demographics (Students).
- **Next:** What if prices change every second? (Dynamic Pricing).



Part 3: Dynamic Pricing & Algorithms

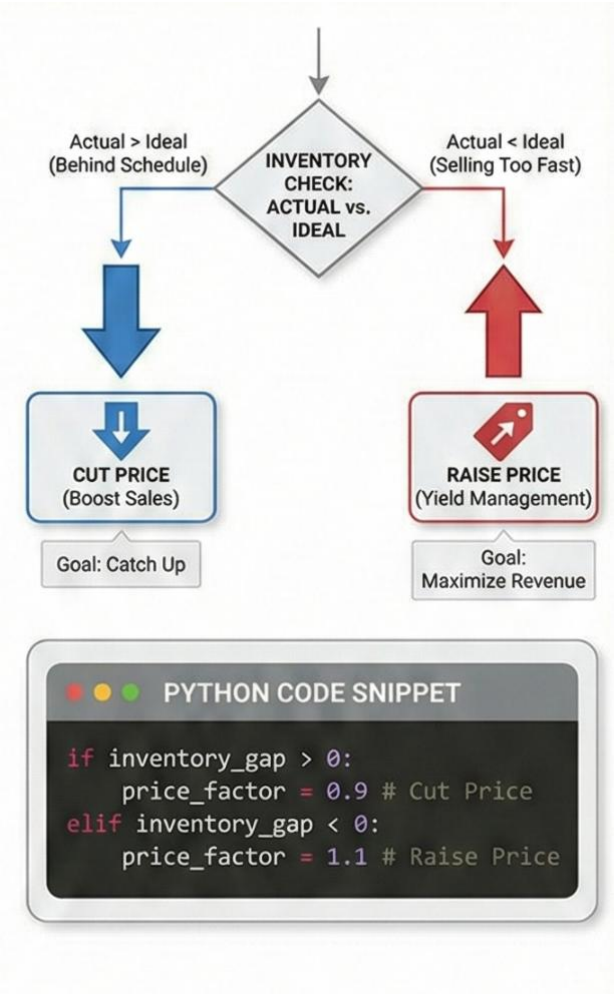
Dynamic Pricing (The Hotel Model)

- **Context:** Perishable Inventory (Hotel rooms, Airline seats).
  - If a room is empty tonight, its value tomorrow is $0.
- **The Goal:** Yield Management. Fill the plane/hotel at the highest possible average price.
- **The Algorithm:**
  - Price ↑ as Inventory ↓. (Scarcity)
  - Price ↓ as Time Remaining ↓. (Expiration)



Simulating Hotel Pricing

- In the Notebook, we build a dynamic_pricing_algorithm.
- **Logic:**
  - Ideal_Inventory = (Time_Left / Total_Time) * Total_Rooms
  - If Actual_Inventory > Ideal: We are behind schedule → **Cut Price**.
  - If Actual_Inventory < Ideal: We are selling too fast → **Raise Price**.



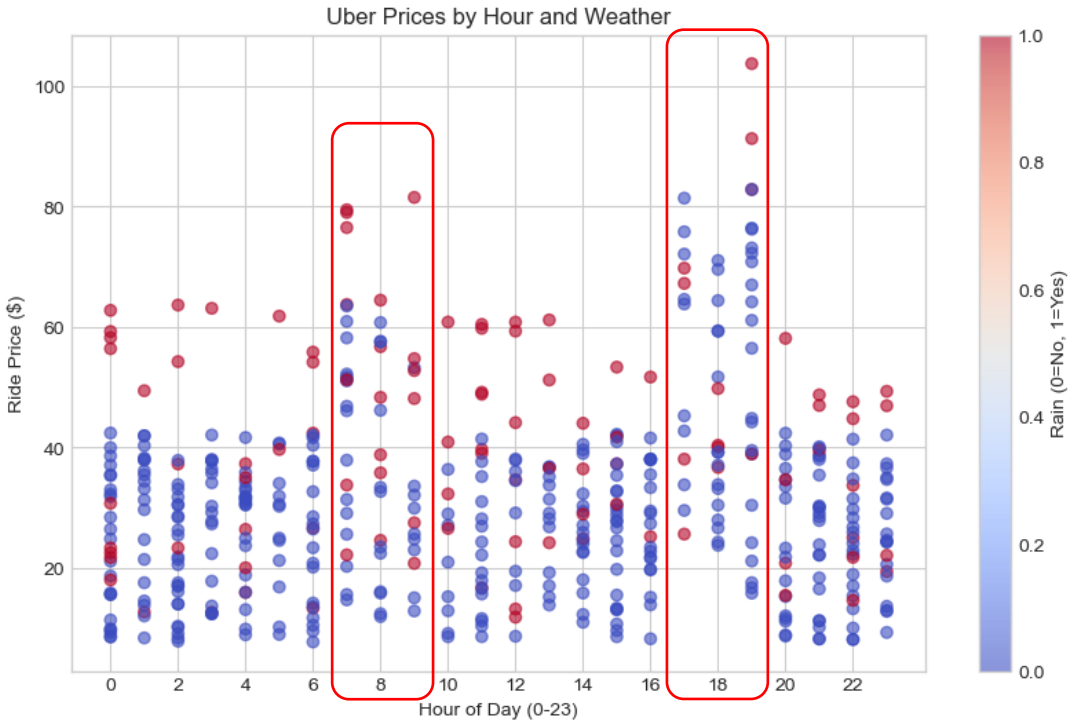
Case Study: Uber Surge Pricing

- **The Problem:** Friday night, raining. Demand > Supply.
- **Static Solution:** First come, first served. Wait times explode. No cars available.
- **Dynamic Solution (Surge):** Price x2.0.
  - **Demand Side:** People who don't *really* need a ride wait (Demand ↓).
  - **Supply Side:** Drivers sitting at home see high fares and get on the road (Supply ↑).
- **Result:** Market clears. You get a car (if you pay).



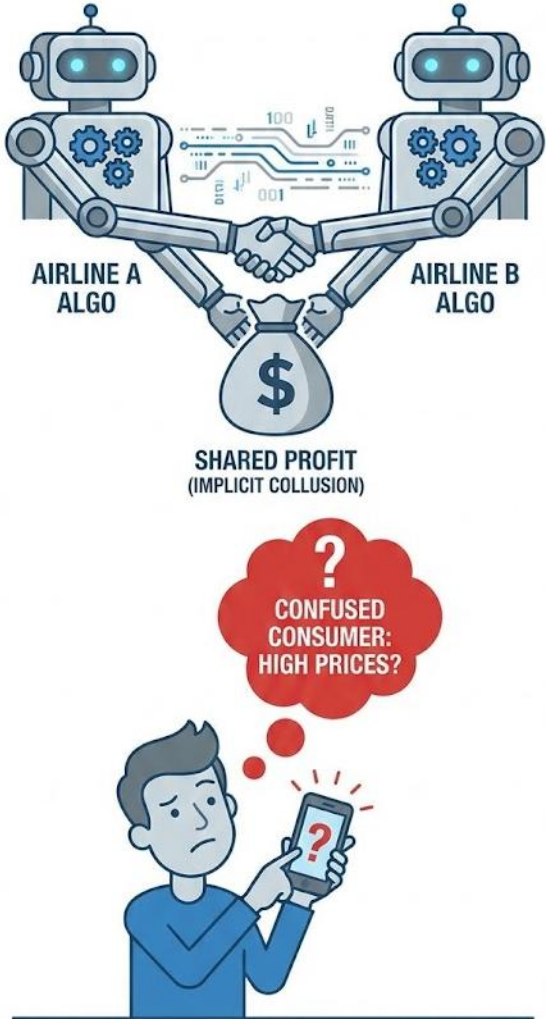
Visualizing Surge Data

- In the tutorial, we analyze real ride data.
- We simulate a "Rain Surge."
- **Code:** `multiplier = np.where(is_raining == 1, 1.5, 1.0)`
- **Graph:** We plot Price vs. Hour. You see spikes during Rush Hour (8am, 6pm) and Rain.



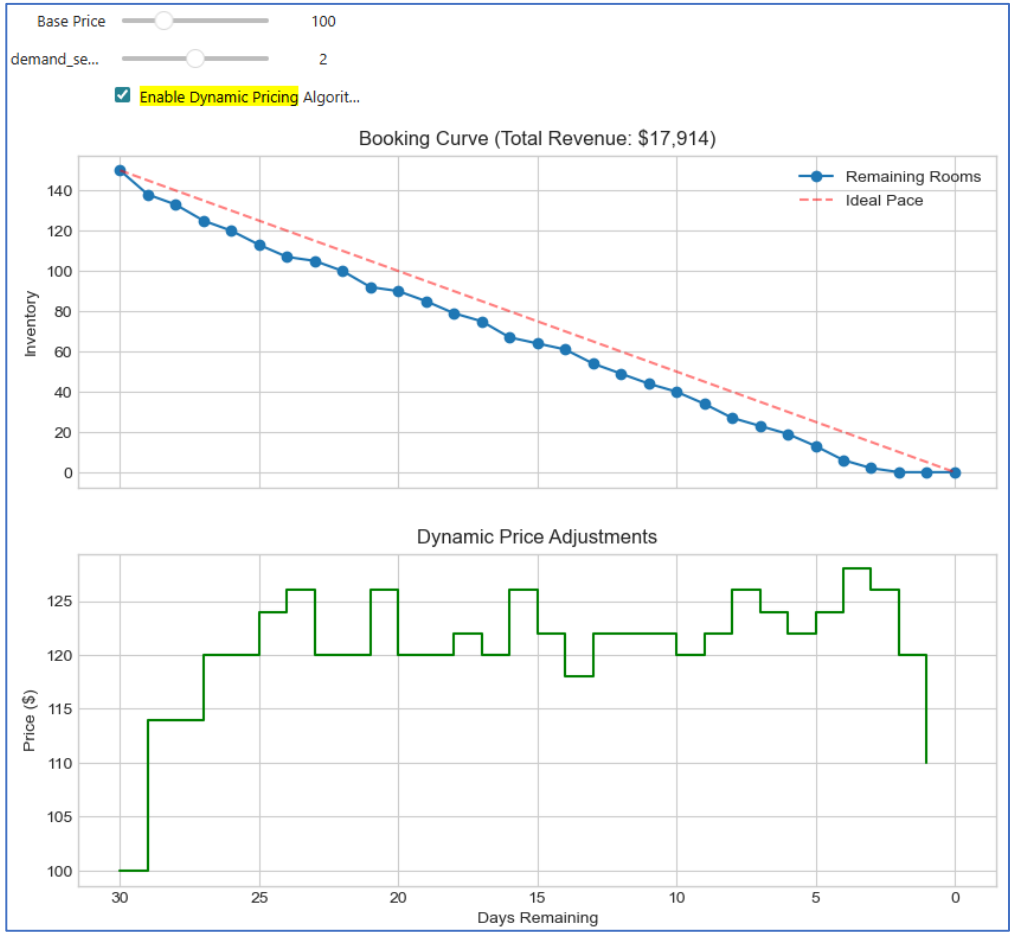
Algorithmic Collusion (The Dark Side)

- What happens if everyone uses the same pricing algorithm?
- **Scenario:** Airline A and Airline B use the same software (e.g., Rainmaker).
- **Outcome:** The algorithms might "learn" that if they both keep prices high, they both profit.
- **Result:** Implicit Collusion (Oligopoly pricing) without humans ever talking.



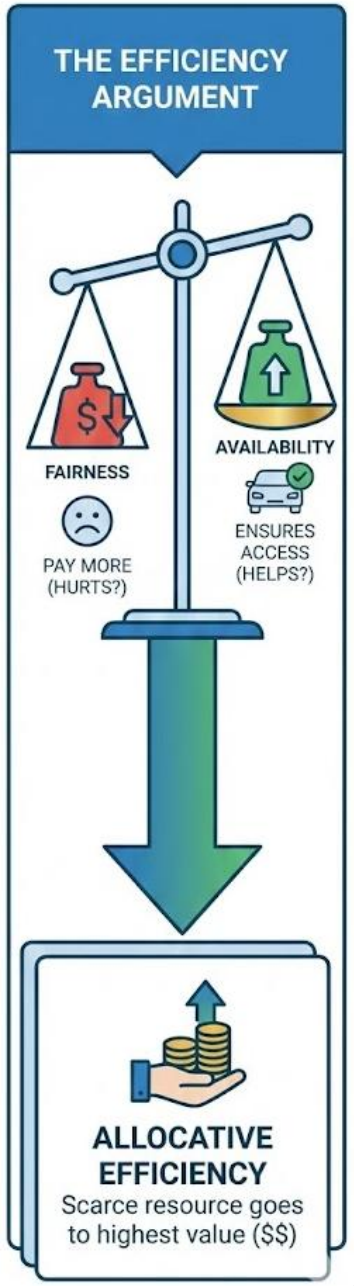
Implementing Dynamic Pricing in Python

- We use ipywidgets to make the simulation interactive.
- **Slider:** Base Price.
- **Checkbox:** Enable Algorithm.
- **Experiment:** Watch what happens to "Revenue" when you turn the algorithm **OFF**. (Spoiler: Revenue drops).



The Efficiency Argument

- Does Dynamic Pricing hurt consumers?
- **Yes:** You pay more during storms.
- **No:** It ensures availability. Without surge, there are NO cars. With surge, there are cars for those who need them most (High WTP).
- **Allocative Efficiency:** The scarce resource goes to the person who values it most ($$).



Part 3 Summary

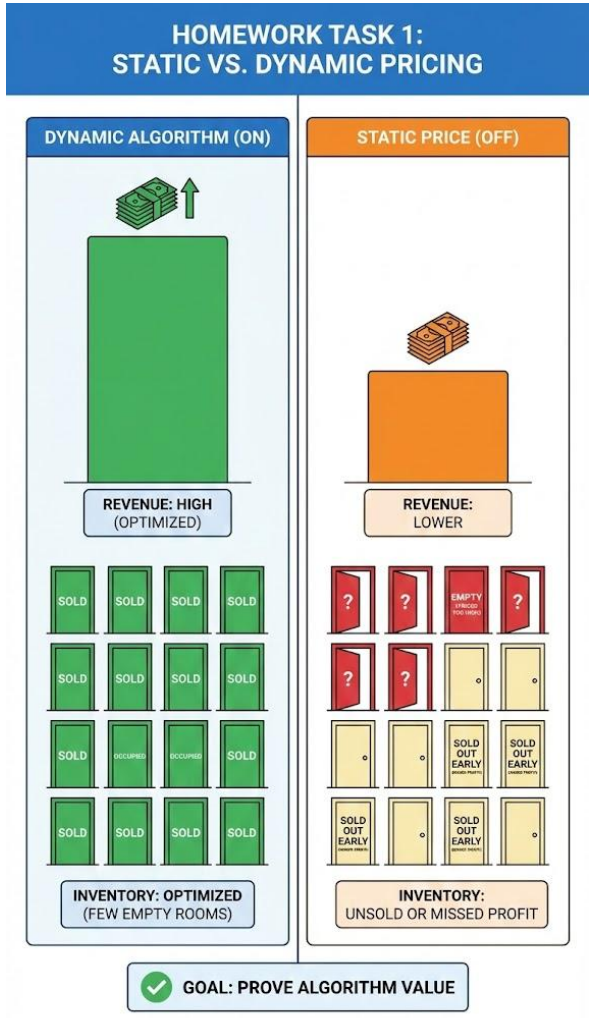
- Dynamic pricing matches Supply and Demand in real-time.
- It prevents shortages (Uber) and spoilage (Empty Hotel Rooms).
- Algorithms monitor "Pace" (Inventory vs. Time).
- **Next:** Applying this to your homework.



Part 4: Homework & Conclusion

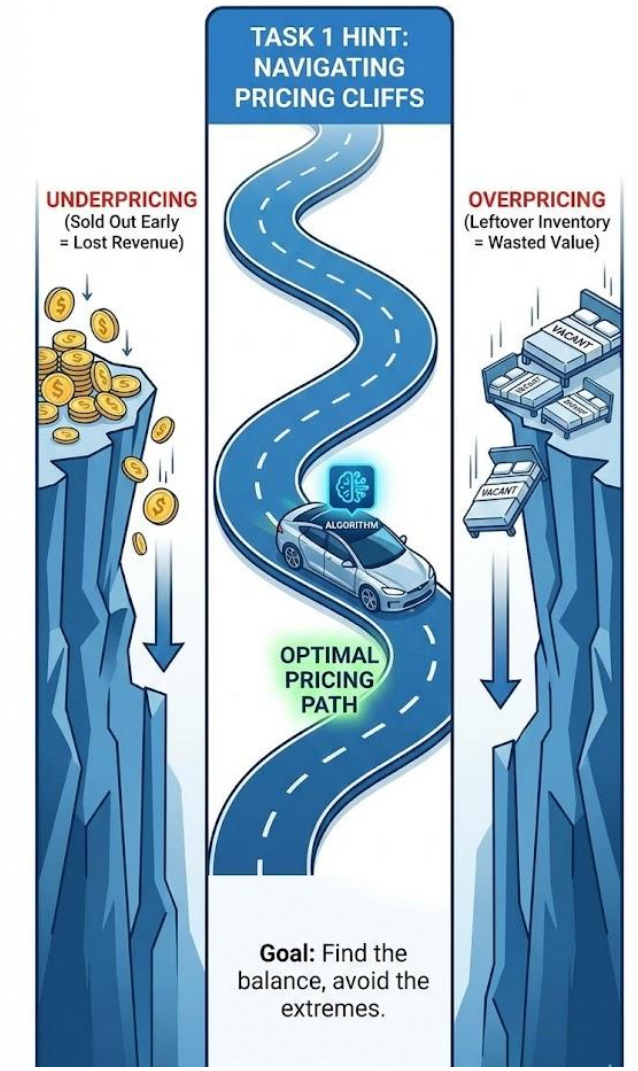
Homework Task 1: Static vs. Dynamic

- **Goal:** Prove the value of the algorithm.
- **Step 1:** Run the Hotel Simulator with the algorithm **ON**. Note the Revenue.
- **Step 2:** Run it with the algorithm **OFF** (Static Price).
- **Question:** What happened to "Unsold Inventory"?
  - *Hint:* Did you price it too high and end up with empty rooms? Or too low and sell out early (missing profit)?



Hint for Task 1

- **Underpricing:** Selling out by Day 15 means you left money on the table. (Price was too low).
- **Overpricing:** Having 50 rooms left at Day 0 means you wasted inventory. (Price was too high).
- The Algorithm navigates between these two cliffs.



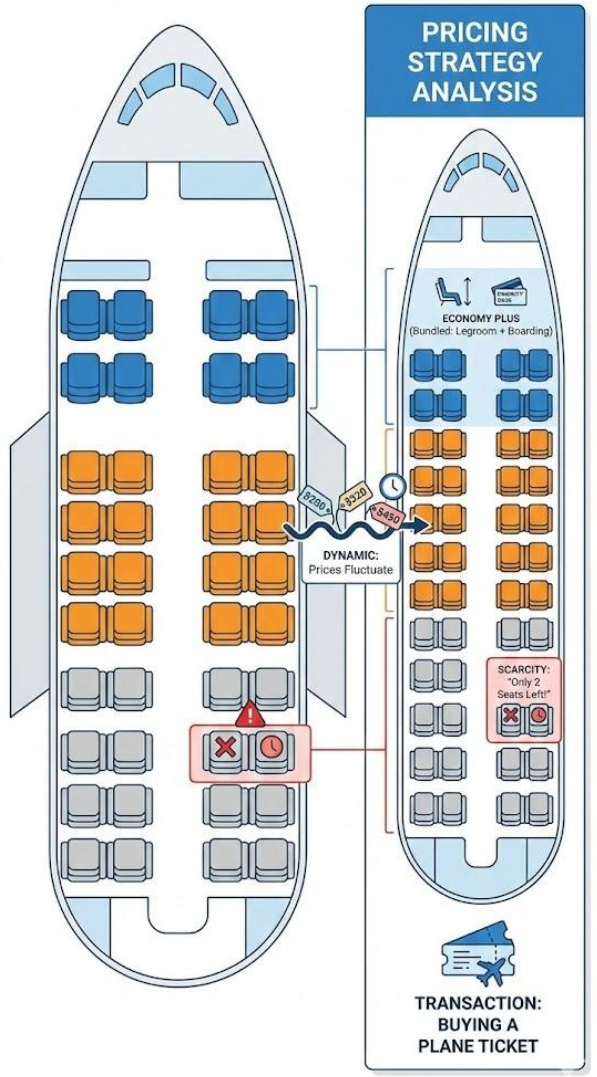
Homework Task 2: Business Transaction Analysis

- **Task:** Analyze a real purchase you made (Coffee, Grab, Shopee, Agoda).
- **Questions:**
  - Did you get a discount? (Price Discrimination).
  - Did the price change? (Dynamic).
  - Was it ฿199? (Psychological).
- **Reflection:** Did this strategy make you buy when you otherwise wouldn't have?



Example Analysis

- **Transaction:** Buying a Plane Ticket.
- **Strategy:**
  - **Dynamic:** Price changed 3 times in a week.
  - **Discrimination:** "Economy Plus" (Bundling legroom + early boarding).
  - **Psychological:** "Only 2 seats left at this price!" (Scarcity).



Final Recap

- **Marketing:** Creating value/demand.
- **Pricing:** Capturing that value.
- **Strategies:** Discrimination (Segmentation), Dynamic (Real-time), Psychological (Behavioral).
- **Tech:** Algorithms now make these decisions milliseconds faster than humans.



Let’s check out our Notebook

